



Varendra University
Department of Computer Science and Engineering
Course Outline

Course Title: Operating System and System Programming Lab

Course Code: CSE 3204

Course Type: Lab

Credits: 1.5

Prerequisite Knowledge: CSE 1202

Year and Semester: 3th Year, Summer semester, 2026

Instructor's details:

Name: Md. Fayzul Islam

Designation: Lecturer

WhatsApp: 01323117174

Office Room: (Ground Floor, East Block)

Contact Email: fayzul@vu.edu.bd

Power Point Slide's and Virtual Learning Platform Web Link: Teams classroom.

Motivation of Course:

Learning Operating Systems and System Programming motivates me to understand how software directly interacts with hardware and manages resources efficiently. It builds a strong foundation for writing high-performance, reliable programs and deepens my problem-solving skills at the system level.

Course Objective:

The objective of this course is to provide a clear understanding of the fundamental concepts of Operating Systems and System Programming. It aims to teach how an operating system manages processes, memory, files, and I/O devices efficiently. The course also focuses on system-level programming concepts such as system calls, process control, and inter-process communication. Students will learn how to design and implement programs that interact directly with the operating system. Finally, the course prepares students to analyze system performance and develop reliable, efficient system software.

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

COs	Description	Taxonomy domain/level	POs	WK	WP	EA	Teaching-Learning strategy	Assessment strategy
CO1	Execute basic Linux commands for file management, process control, and system monitoring	Psychomotor/ Manipulation	2	2			Hands on Class	Continuous Assessment
CO2	Evaluate the performance of different process scheduling algorithms and memory management techniques through implementation and analysis	Cognitive/ Evaluate	5	3	1,2,3	1,3	Hands on Class	Lab Final

Teaching-Learning strategy

COs	Teaching-Learning strategy	Assessment strategy
CO1	Discussion and hands on practice in lab	Continuous Assessment
CO2	Discussion and hands on practice in lab	Lab Final + Quiz

Assessment Detail

Assessment Tools		Marks (%)	
Continuous Assessment (CA)	Class Participation	10%	50%
	Continuous Assessment (Experiment, Report, Viva)	40%	
Summative Assessment (SA)	Final Examination (Lab Test, Quiz, Viva, Project)	50%	50%
Total		100%	

Textbook:

1. A. Silberschatz and Peter Baer Galvin : **Operating System Concepts**, Wiley Publisher
2. Tanenbaum : **Operating Systems**, Prentice Hall

Books Recommended:

1. Madnick and J. Donovan : **Operating Systems**, McGraw-Hill

Course Conducting/Course Policies:

1. It is the student's responsibility to gather information about the assignments and covered topics if he/she does miss the lecture
2. Regular class attendance is mandatory. Points will be taken off for missing classes.
3. Without **50% of** attendance, sitting for final exam is **NOT allowed**.
4. The students must enter the **classroom in time** to get the attendance. **No student** will be allowed to enter the classroom after the attendance has been done.
5. Once the attendance is done, a student can leave the class if he or she thinks that he or she is not getting benefits from the class
6. The reading materials for each class will be available at the **above given link** prior to that class so that student may have a cursory look into the materials.
7. The date and syllabus of quiz/class test will be announced in time in Microsoft Team
8. Students will be **notified** in due time for class cancelation, extra class, make-up class and tutorial class.
9. Students are encouraged to participate in the class discussion and to **ask questions**. The student can ask any question without any hesitation as long as he or she can't understand the topics being discussed; please keep in mind that if you don't understand, it's not your fault, it's my limitation that I could not make you understand. The class is expected to be interactive. The class is expected to be **interactive**.
10. It is expected that the student will also provide some new knowledge related to the curriculum and then make the class as a **place of knowledge sharing among all participants, both teacher and students**.
11. Any attempt for **unfairmeans** in the examination is **strictly prohibited**.

Class & Exam Schedule, Topics and Readings:

Sessions	Topics	Readings
Week-1	Introduction to Linux & Ubuntu Environment	Discussion and hands on practice in lab
Week-2	File and Directory Management Commands	Discussion and hands on practice in lab
Week-3	File Viewing, Editing & Permissions, Shell Scripting Basics	Discussion and hands on practice in lab

Week-4	Lab assessment (from lab week 1-3)	
Week-5	Process Management & System Monitoring	Discussion and hands on practice in lab
Week-6	Process Management & System Monitoring.	Discussion and hands on practice in lab
Week-7	Mid Lab Test	
Week-8	CPU Scheduling Algorithms	Discussion and hands on practice in lab
Week-9	Preemptive and non-preemptive, CPU Scheduling Algorithms	Discussion and hands on practice in lab
Week-10	Memory Management Techniques	Discussion and hands on practice in lab
Week-11	Memory Management Techniques	Discussion and hands on practice in lab
Week-12	Page Replacement Algorithms	Discussion and hands on practice in lab
Week-13	Performance Evaluation & Lab Final	Discussion and hands on practice in lab
Week-14	To be Declared	